

A photograph of a wind farm with several white wind turbines on a grassy hill. The foreground is a golden-brown field with several hay bales. The background shows a dense forest and a body of water under a clear sky.

Phase 3 – Wind Turbine Design

Pingfan Teng

S2078106

22/04/2024

A photograph of a wind farm with several white wind turbines on a grassy hill. In the foreground, there are golden-brown hay bales. The background shows a dense forest and a body of water under a clear sky.

Final Design parameters

Rated power: 4.8MW

Rated wind speed: 12 m/s

Rotor diameter: 113.3 m

Tower height: 78.8 m

A photograph of a wind farm with several white wind turbines on a grassy hill. The sky is blue with some clouds, and the foreground is a field of dry grass with some hay bales.

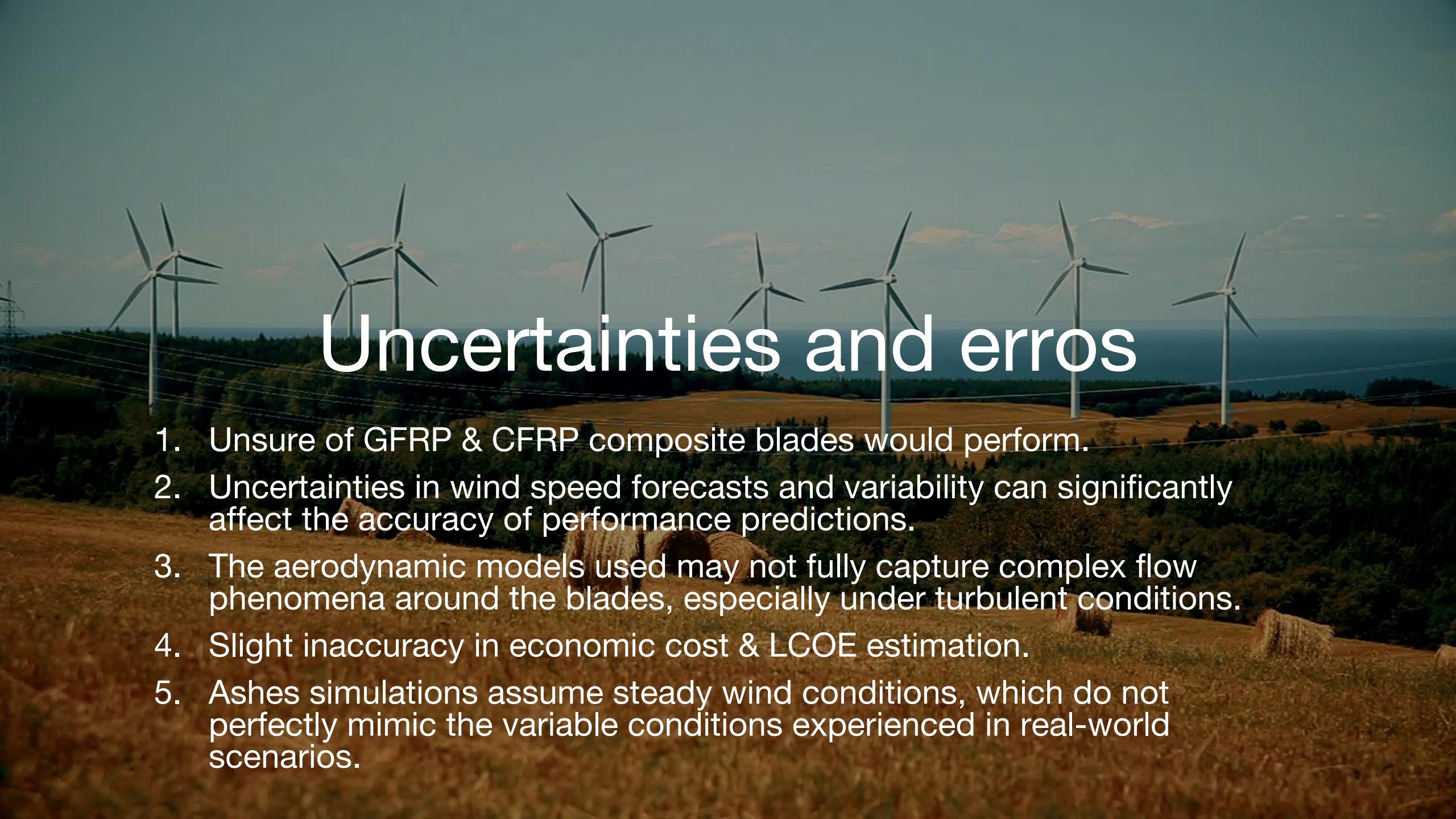
Assumptions

1. Air density = 1.225 kg/m^3
2. Maximum tip speed of 80 m/s
3. Turbulent wind, vibrations or motions of the tower, rotating rotor with yaw/tilt/cone angle.
4. Economic cost LCOE etc (please refer to the spreadsheet for detailed assumptions)
5. Steady wind speed (Ashes)



Accomplished Tasks

1. Created the designed wind turbine model on Ashes.
2. Analyzed the aerodynamics & loading effects on Ashes (from design model).
3. Analyzed blade structural integrity and optimized the blades material.
4. Analyzed the dynamic response of the designed model.
5. Estimated the economic cost and LCOE of the wind farm project.

A photograph of a wind farm with several white wind turbines on a grassy hill. The sky is blue with some clouds, and the foreground shows dry grass and some hay bales.

Uncertainties and erros

1. Unsure of GFRP & CFRP composite blades would perform.
2. Uncertainties in wind speed forecasts and variability can significantly affect the accuracy of performance predictions.
3. The aerodynamic models used may not fully capture complex flow phenomena around the blades, especially under turbulent conditions.
4. Slight inaccuracy in economic cost & LCOE estimation.
5. Ashes simulations assume steady wind conditions, which do not perfectly mimic the variable conditions experienced in real-world scenarios.

A photograph of a wind farm with several white wind turbines on a grassy hill. In the foreground, there are several large, round hay bales. The background shows a blue sky with some clouds and a distant horizon.

Recommendations

1. Recommended to further research on the GFRP & CFRP composite blades, by computing the accurate mass and re-constructing the Campbell diagram.
2. Recommended to further investigate the structural loading dynamically, instead of static.
3. Recommend to further research and develop a better composite material for the blades, either lightweight and avoids resonance.

The background image shows a series of white wind turbines standing on a grassy hill. In the distance, a blue body of water, likely the ocean, is visible under a clear sky. The foreground consists of dry, golden-brown grass.

What is next?

1. Complete a detailed life cycle assessment of the project.
2. Complete a detailed carbon assessment of the project.
3. Start the layout of the electrical system (power cables selection, grid connection, substation lay out etc.).
4. Gather local community and update the project progression.
5. Start purchasing necessary licenses.
6. Ensure legal compliances during the manufacturing and construction stage.

A landscape photograph featuring a row of white wind turbines on a distant ridge. The foreground is a golden-brown field with several large hay bales. A dense green forest separates the field from the turbines. The sky is a pale blue with light clouds. The text "Thank you for your patience" is centered in white.

Thank you for your patience